

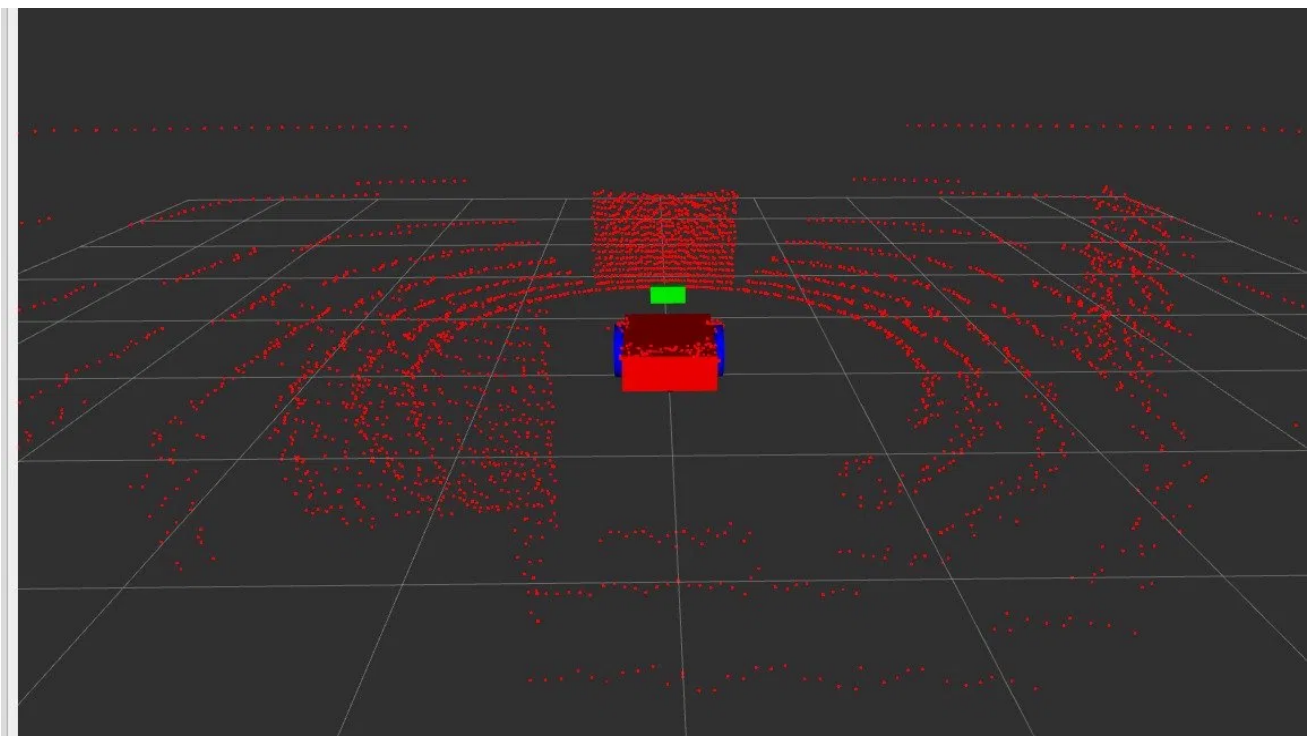
LiDAR Sensor Simulation and Performance Report

This report presents the simulation results and comparative analysis of two LiDAR sensors configured in a ROS 2 + Gazebo Harmonic simulation environment. The sensors evaluated are:

- 1. Ouster OS1-64
- 2. Velodyne VLP-16

The analysis includes point cloud visualization, topic publishing frequency, PointCloud2 message structure, and TF transformation verification.

1. Ouster OS1-64 LiDAR Results



The above image shows the point cloud generated by the simulated Ouster OS1-64 LiDAR sensor. The sensor produces a dense 64-channel point cloud with improved environmental coverage and higher spatial resolution compared to lower beam-count sensors.

ROS 2 Topic Frequency Analysis (/scan_3d)

Average observed publishing rate ranged between approximately **8.4 Hz to 9.7 Hz**. The topic frequency remained stable for most of the simulation duration with minor deviations.

Metric	Observed Value
Average Frequency	8.4 Hz – 9.7 Hz
Minimum Interval	0.089 s
Maximum Interval	0.502 s

Standard Deviation	0.052 – 0.066 s
Topic Name	/scan_3d

PointCloud2 Message Structure

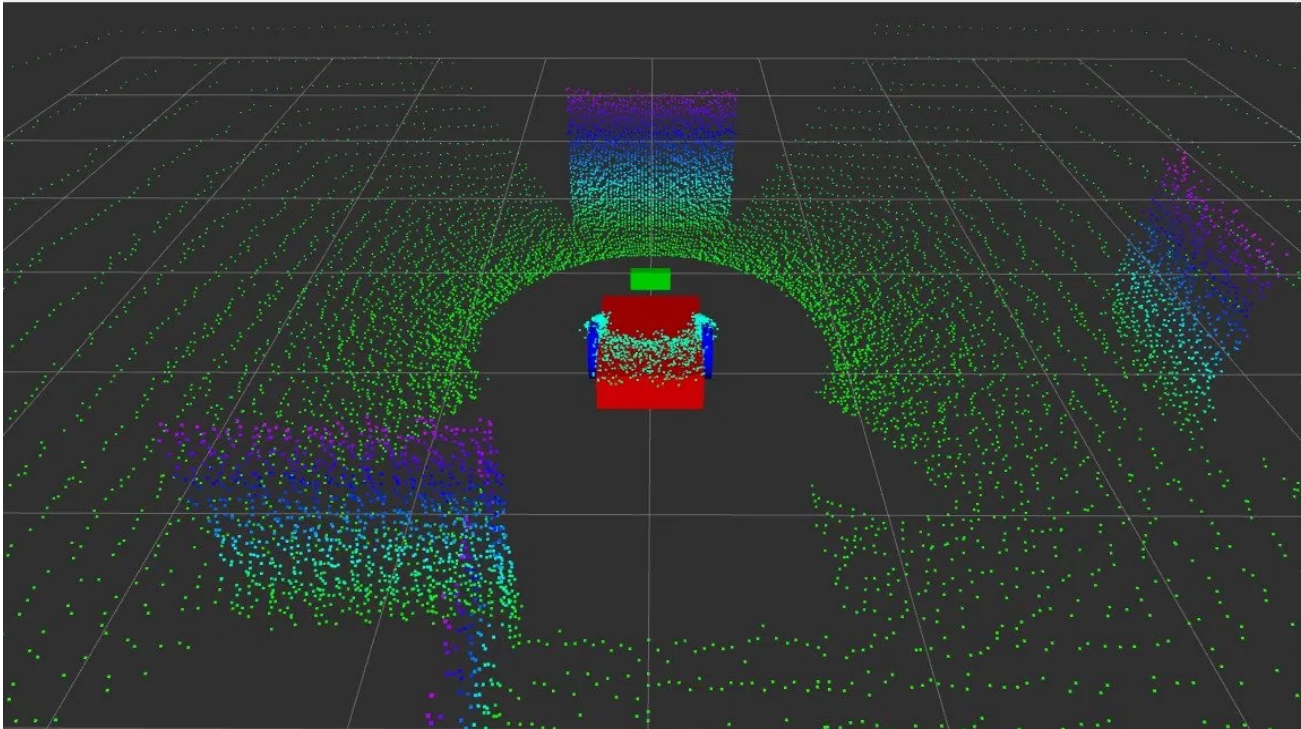
Parameter	Value
Height	64
Width	360
Point Step	32 bytes
Row Step	11520 bytes

TF Transform Verification

The TF transformation between **base_link** and **lidar_link** was successfully verified. The sensor frame was positioned relative to the robot base using the following transform:

Transform Parameter	Value
Translation X	0.320 m
Translation Y	-0.405 m
Translation Z	0.000 m
Rotation (RPY)	[0.0, 0.0, 0.0]

2. Velodyne VLP-16 LiDAR Results



The above image shows the point cloud generated by the simulated Velodyne VLP-16 LiDAR sensor. Compared to the OS1-64, the VLP-16 produces a sparser point cloud due to the lower number of vertical channels. However, it remains computationally lighter and suitable for many robotic perception and navigation applications.

3. Comparative Analysis

Feature	OS1-64	VLP-16
Vertical Channels	64	16
Point Cloud Density	High	Moderate
Environmental Detail	Very Detailed	Moderate
Computation Cost	Higher	Lower
Suitability	Advanced Perception	Efficient Navigation

4. Conclusion

The simulation successfully demonstrated the integration and visualization of both the Ouster OS1-64 and Velodyne VLP-16 LiDAR sensors in ROS 2 and Gazebo Harmonic. The OS1-64 provided significantly denser and richer point cloud data, while the VLP-16 provided a computationally efficient alternative for navigation and obstacle detection tasks.

ROS 2 topic publishing, TF transformations, and PointCloud2 message structures were successfully validated during testing.